

Notes: Naaraayanan and Wolfenzon (RFS, 2024)

Business Group Spillovers

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1 Big picture

- **Question.** Do business groups affect the investment behavior of *non-affiliated* firms in the local economy, and if so through what mechanism?
- **Core setting.** The paper studies India, where business groups are common and where the Golden Quadrilateral (GQ) highway upgrade created a large positive shock to firms' investment opportunities.
- **Main empirical comparison.** Compare *standalone* firms across regions after the same investment-opportunity shock, asking whether standalones in areas with high business-group prevalence invest less than standalones in areas with low business-group prevalence.
- **Main result.** Yes. Standalone firms' investment sensitivity to the GQ shock is substantially lower in places where business groups are more prevalent.
- **Economic interpretation.** The paper argues that business groups create a spillover onto non-affiliated firms. The leading mechanism is a financing channel: when investment opportunities improve, banks tilt credit toward group-affiliated firms and away from standalones.
- **Why the design is clever.** The paper does *not* compare group affiliates to standalone firms directly. Instead, it compares standalone firms in high- versus low-business-group-share regions around a plausibly common shock. That sidesteps the classic selection problem that organizational form is endogenous.
- **Why the paper matters.** A large literature studies whether internal capital markets make business groups efficient *within the group*. This paper asks a broader equilibrium question: what do groups do to outsiders? The answer is that they can crowd out standalone firms even when aggregate investment in the region does not fall.

2 Data and measurement

2.1 Golden Quadrilateral as the investment-opportunity shock

- **Project.** The GQ upgraded roughly 5,800 kilometers of highway connecting Delhi, Mumbai, Chennai, and Kolkata.
- **Why it matters economically.** The upgrade improved connectivity, lowered transportation costs, improved input sourcing and inventory efficiency, and increased access to regional product markets.
- **Timing variation.** Different highway stretches began construction at different dates. The paper exploits this staggered timing across cities on the eventual GQ route.
- **Useful institutional detail.** Funding came mainly from the federal government and multilateral agencies, so the road construction itself did not directly absorb local bank credit. That matters because the later lending results are easier to interpret as reallocation toward business groups rather than road-financing crowd out.
- **Implementation.** Firms at the two ends of each upgraded stretch are coded as treated at the start of construction. The control group is firms in cities that eventually get GQ upgrades but have not yet been reached.

2.2 Firm-level data

- **Source.** CMIE Prowess, which contains annual financial statements for both listed and private Indian firms.
- **Coverage.** The raw panel runs from 1989 to 2016.
- **Business-group status.** The paper uses Prowess’ classification of whether a firm is affiliated with a business group. That classification is standard in the India business-group literature.
- **Key outcomes.** Investment, profitability, cash flow, debt, total factor productivity, and days sales of inventory.
- **Granularity.** Prowess also reports product-level revenues for manufacturing firms. The paper uses this detail later to study reserved-product industries and rival-opportunity mechanisms.

2.3 Regional banking and loan-level data

- **District-level credit.** From the Reserve Bank of India, used to test whether aggregate bank lending rises after GQ and whether that increase differs across high- and low-business-group-share regions.
- **Loan-level data.** Hand-collected from the Ministry of Corporate Affairs’ credit registry for secured loans. This is crucial for the bank-lending mechanism because it allows the paper to compare the same firm’s borrowing across multiple lenders.
- **Why the loan-level data are powerful.** With firm fixed effects at the borrower level, the paper can absorb firm-level credit demand and identify whether banks with stronger preexisting exposure to group affiliates cut lending to standalones.

2.4 Key variable construction

$$\text{Investment}_{it} = \frac{\text{Net capital expenditure}_{it}}{\text{Total assets}_{i,t-1}}. \quad (1)$$

- **Interpretation.** This is the main firm-level investment rate used throughout the paper.

$$\begin{aligned} \text{BGS}_c &= \frac{\text{Assets of group-affiliated firms in city } c}{\text{Assets of all firms in city } c}, \\ \text{HighBGS}_c &= \mathbb{I}\{\text{BGS}_c \text{ is in the top quartile}\}. \end{aligned} \quad (2)$$

- **Interpretation.** This is the paper’s baseline measure of local business-group prevalence.
- **Economic meaning.** A high value of HighBGS means the local corporate landscape is dominated by business groups.

2.5 Sample construction and descriptive facts

- **Exclusions.** The paper drops financial firms, government-owned firms, observations with implausible leverage or investment ratios, certain industries directly tied to road construction, and very small firms with potentially noisy accounts.

- **Final size.** Table 1 reports 24,709 firm-year observations for all firms and 15,842 for standalone firms in the main sample.
- **Standalone versus average firm.** Standalones are smaller and younger on average than the full sample, but look broadly similar on cash flow, profitability, investment, and debt.

3 Models and empirical specifications

3.1 First step: verify that GQ is a positive shock to investment opportunities

$$y_{ijcst} = \alpha_i + \beta \text{PostGQ}_{ct} + \omega_{jt} + \theta_{st} + \eta_{ijcst}. \quad (3)$$

- **Dependent variable.** Depending on the column, the left-hand side is total factor productivity, days sales of inventory, or investment.
- **Treatment indicator.** PostGQ_{ct} equals 1 in the year of and after the commencement of the GQ upgrade in city c .
- **Fixed effects.** Firm fixed effects, industry-year fixed effects, and state-year fixed effects.
- **Interpretation of β .** It captures how firms change once their city gets the highway upgrade, relative to firms on the eventual GQ route that have not yet been upgraded.
- **Identification logic.** Because the comparison is only across cities that eventually receive the highway, the paper uses timing rather than treated-versus-never-treated level differences.
- **Expected signs.** If GQ improves investment opportunities, β should be positive for productivity and investment and negative for days sales of inventory.

3.2 Main spillover specification

$$y_{ijcst} = \alpha_i + \beta_1 \text{PostGQ}_{ct} + \beta_2 (\text{PostGQ}_{ct} \times \text{HighBGS}_c) + \omega_{jt} + \mu_{jt} \times \text{HighBGS}_c + \theta_{st} + \gamma_{st} \times \text{HighBGS}_c + \varepsilon_{ijcst}. \quad (4)$$

- **Dependent variable.** In the main tests, y_{ijcst} is the investment rate of standalone firms.
- **Coefficient of interest.** β_2 . A negative β_2 means that standalone firms invest less after GQ when they are located in high-business-group-share regions.
- **Why the interacted fixed effects matter.** The paper lets industry-year shocks and state-year shocks differ between high- and low-business-group-share places. That is a strong way to absorb alternative regional explanations.
- **Economic interpretation.** This is not asking whether group firms invest more than standalone firms. It asks whether the *same type of firm* — a standalone — responds differently to a common opportunity shock depending on the local prevalence of business groups.

3.3 Dynamic event-study version

$$\begin{aligned} \text{Investment}_{ijcst} = & \alpha_i + \sum_{k=-4}^{-1} \mu_k D_{ct}^k + \sum_{k=1}^{10} \mu_k D_{ct}^k \\ & + \sum_{k=-4}^{-1} \lambda_k \left(D_{ct}^k \times \text{HighBGS}_c \right) + \sum_{k=1}^{10} \lambda_k \left(D_{ct}^k \times \text{HighBGS}_c \right) \\ & + \omega_{jt} + \vartheta_{jt} \times \text{HighBGS}_c + u_{ijcst}. \end{aligned} \quad (\text{ES})$$

- **Interpretation.** D_{ct}^k is an event-time dummy for year k relative to highway commencement. The coefficients λ_k trace how the investment gap between high- and low-business-group-share regions evolves before and after treatment.
- **Why it matters.** The pre-treatment λ_k coefficients are a visual pretrend test. The post-treatment coefficients show persistence of the spillover.

3.4 Loan-supply mechanism specification

$$\Delta \text{Loan}_{ib} = \alpha_i + \beta_1 \text{GroupExposure}_b + \eta_{ib}. \quad (5)$$

$$\text{GroupExposure}_b = \text{bank } b\text{'s pre-GQ lending to group-affiliated firms.} \quad (6)$$

- **Left-hand side.** Change in the average loan amount to a standalone firm from bank b , comparing the 5 years after GQ with the 5 years before GQ.
- **Firm fixed effects.** α_i absorb the borrower's credit demand. So identification comes from how different banks change lending to the *same* standalone firm.
- **Interpretation of β_1 .** If $\beta_1 < 0$, banks more exposed to business groups cut lending more to standalone firms after the shock.
- **Why this is the cleanest mechanism test in the paper.** It gets closest to isolating credit supply rather than credit demand.

4 Identification

4.1 This is a quasi-experimental difference-in-differences design, not an IV paper

- There is no external instrument. The key source of variation is the staggered timing of GQ construction across cities on the eventual highway route.
- The paper compares cities that all eventually receive the highway, which avoids a crude comparison between highway and non-highway places.
- The core assumption is that timing of construction is orthogonal to changes in firms' investment opportunities, after controlling for firm, industry-year, and state-year effects.

4.2 Why the design is plausible

- **Institutional rollout.** Delays were driven by land acquisition, utility shifting, environmental approvals, contractor cash-flow problems, and law-and-order issues. Those reasons make timing plausibly noisy from the perspective of firm investment demand.
- **Within-route comparison.** The control group is “yet-to-be-treated” GQ cities, which makes treated and control cities much more comparable than a never-treated comparison.
- **Validation step.** Before moving to spillovers, the paper shows that GQ actually raises TFP, improves inventory efficiency, and increases investment. So the treatment behaves like a positive opportunity shock.

4.3 Main threats and how the paper handles them

- **Threat 1: Standalone firms differ across high- and low-business-group-share regions.** The paper compares pre-GQ firm characteristics and finds broad similarity; it also runs “horse race” regressions controlling for other regional characteristics such as listing share, firm age, and TFP.
- **Threat 2: Regions differ in other ways.** The paper compares bank-branch density, the fraction of listed firms, road quality, and labor-market conditions, and finds little evidence of systematic differences explaining the main result.
- **Threat 3: The GQ shock might be stronger in some regions.** The paper shows ex ante road quality is similar across high- and low-business-group-share areas, ex post road engineering standards are similar, stock-price reactions to new-plant announcements are similarly positive across regions, and inventory-efficiency improvements are also similar.
- **Threat 4: Two-way fixed effects with staggered adoption.** In the appendix, the paper checks robustness with the Borusyak-Jaravel-Spiess imputation estimator and reaches similar conclusions.
- **Threat 5: Exit, M&A, or entry composition effects.** Appendix results show the main coefficient survives dropping likely exits and is not explained by differential mergers or firm entry.

4.4 What the identification does and does not establish

- **Strong point.** The paper credibly establishes that standalone investment responds less to the same broad shock in regions dominated by business groups.
- **More limited point.** The mechanism evidence is persuasive but not as airtight as the reduced-form result. In particular, labor-market crowd out cannot be tested directly with the available data.
- **Welfare limitation.** The paper documents spillovers, but it does not prove that aggregate allocation is worse. If group affiliates are more productive, reallocation toward them could in principle be efficient.

5 Tables and figures

5.1 Figure 1: Temporal variation in GQ construction

Read:

- The start dates of road-construction subsegments are spread over time, with the bulk of activity concentrated between 2000 and 2006.
- The figure matters because the paper's design relies on staggered timing rather than a single aggregate before-after break.
- The four major corridor segments look broadly similar in timing, which helps support the idea that no one corridor is driving the results.

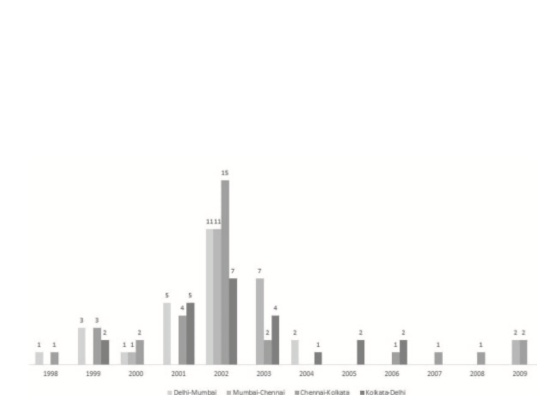


Figure 1
Temporal variation in GQ construction
This figure illustrates the temporal variation in the commencement of construction of the four segments forming part of the Golden Quadrilateral, which connects the four nodal cities of Delhi, Mumbai, Chennai, and Kolkata. The height of each bar corresponds to the number of subsegments that began construction each year. Data source: National Highway Authority of India.



Figure 2
GQ construction evolution over time
This figure illustrates the spatial variation of segments at different points in time along the GQ network. The network is part of the 5,846 kilometer stretch of the GQ connecting the four nodal cities of Delhi, Mumbai, Chennai, and Kolkata. Map source: National Highway Authority of India.

5.2 Figure 2: Spatial evolution of GQ construction

Read:

- The map shows how the road network expands over time around the quadrilateral connecting the four major cities.
- The visual makes clear that treatment is geographically broad and not concentrated in one narrow local cluster.
- Economically, it reinforces that the shock is about improved connectivity and market access rather than one isolated local public-works project.

5.3 Table 1: Summary statistics

Read:

- The average firm is much larger than the average standalone firm: mean assets are INR 8,176 million for all firms versus INR 2,593 million for standalones.
- Standalones are also younger on average: 27 years versus 30 years in the full sample.
- But standalones look broadly similar on operating measures: investment is 0.39 versus 0.38, profitability is 0.12 in both samples, and debt is only slightly lower.
- The takeaway is that standalones are smaller, but not obviously worse on the key firm-level observables used later in the paper.

Table 1
Summary statistics

A. All firms						
	N	Mean	SD	P25	P50	P75
Assets (INR millions)	24,709	8,176	73,751	290	919	3,075
Firm age (years)	24,709	30	21	16	23	35
Cash flow	24,709	0.06	0.11	0.00	0.06	0.12
Profitability	24,709	0.12	0.10	0.07	0.11	0.16
Listed	24,709	0.63	0.48	0.00	1.00	1.00
Investment	24,709	0.38	0.22	0.21	0.35	0.51
Debt	23,755	0.27	0.27	0.09	0.22	0.36
Total factor productivity	21,494	3.10	2.57	1.80	2.39	3.31

B. Standalone firms						
	N	Mean	SD	P25	P50	P75
Assets (INR millions)	15,842	2,593	11,873	236	657	1,893
Firm age (years)	15,842	27	20	16	22	31
Cash flow	15,842	0.05	0.11	0.00	0.05	0.11
Profitability	15,842	0.12	0.10	0.07	0.11	0.16
Listed	15,842	0.57	0.50	0.00	1.00	1.00
Investment	15,842	0.39	0.22	0.22	0.36	0.53
Debt	15,842	0.25	0.19	0.10	0.23	0.36
Total factor productivity	10,308	3.19	2.50	1.91	2.50	3.45

This table reports the descriptive statistics of firm characteristics for our sample. Panel A reports the descriptive statistics for all firms, while panel B reports the statistics for standalone firms. From the overall sample of firms in Prowess for the period starting from 1989 to 2016, we exclude all financial firms, firms owned by central and state governments, and firms with less than 3 years of data with positive values of total assets and sales, and drop observations with a ratio of investment to lagged total assets and leverage greater than 1. Further, we keep firm-year observations with nonmissing values for assets, firm age, cash flow, and profitability. All the financial variables are adjusted for inflation using the Wholesale Price Index at 2010 constant prices. We also correct for changes in the financial reporting year by adjusting values for the number of months. To mitigate the effect of outliers, we winsorize ratios at 1% tails and total factor productivity at 2.5% tails. All variables are defined in [Internet Appendix Table 14](#). Data source: CMIE Prowess.

Table 2
Productive efficiency and investment around GQ upgrades

Dependent variable	TFP	Days sales of inventory	Investment
	(1)	(2)	(3)
PostGQ	0.146** (0.067)	-3.961* (2.350)	0.026** (0.012)
Fixed effects:			
Firm	Yes	Yes	Yes
Industry $\times$ year	Yes	Yes	Yes
State $\times$ year	Yes	Yes	Yes
Adjusted- R^2	0.79	0.63	0.70
Observations	21,494	21,053	24,709
Sample: All firms	Yes	Yes	Yes

This table reports changes in productive efficiency and investment for firms located along GQ around the upgrade of the road network. The dependent variable in column 1 is *TFP*, the total factor productivity, which is estimated using the methodology outlined in [Levinsohn and Petrin \(2003\)](#). The dependent variable in column 2 is *Days sales of inventory*, defined as the ratio of ending inventory to cost of goods sold multiplied by 365, and the dependent variable in column 3 is *Investment*, defined as the net capital expenditure divided by lagged total assets. *PostGQ* is an indicator variable taking the value of 1 for all years, including and after the GQ upgrade in the city. All regressions include firm, industry-year, and state-year fixed effects. Note that in column 1, TFP is estimated for manufacturing firms, and in column 2, we restrict the sample to firms with days sales of inventory between 5 and 150 days to mitigate the effect of outliers. Standard errors, reported in parentheses, are corrected for heteroscedasticity and autocorrelation and clustered at the city level. All variables are defined in [Internet Appendix Table 14](#). * $p < .1$; ** $p < .05$; *** $p < .01$. Data source: CMIE Prowess.

5.4 Table 2: Productive efficiency and investment around GQ upgrades

Read:

- **TFP rises.** The coefficient on PostGQ is 0.146 and statistically significant.
- **Inventory efficiency improves.** Days sales of inventory fall by about 3.96 days.
- **Investment rises.** The investment coefficient is 0.026, which the paper interprets as about a 6% increase relative to the average firm.
- This is the validation table for the entire research design: GQ really behaves like a positive shock to firms' investment opportunities.

5.5 Table 3: Baseline spillover result

Read:

- For standalone firms, the coefficient on PostGQ $\times$ HighBGS is -0.038 in the preferred specification and highly significant.
- The coefficient on PostGQ is about 0.039, so low-business-group-share regions see a positive investment response while high-business-group-share regions see roughly no net increase.
- Relative to the average standalone investment rate, this is about a 10% decline in investment sensitivity.

- This is the paper’s central reduced-form result: business groups depress the responsiveness of standalones to new opportunities.

Table 3
Business group prevalence and standalone investment

Dependent variable	Investment	
	(1)	(2)
<i>PostGQ</i>	0.038 (0.029)	0.039 (0.028)
<i>PostGQ</i> × <i>High BGS</i>	−0.012*** (0.000)	−0.038*** (0.012)
Fixed effects:		
Firm	Yes	Yes
High BGS × industry × year	No	Yes
High BGS × state × year	Yes	Yes
Adjusted- R^2	0.70	0.70
Observations	15,842	15,842
Sample: Standalone firms	Yes	Yes

This table reports estimates from regressions relating business group prevalence to standalone firms’ investment around GQ upgrades. *PostGQ* is an indicator variable taking the value of 1 for all years including and after the GQ upgrade in the city. *High BGS* is an indicator variable set to 1 if the share of assets of group-affiliated firms from that city is in the top quartile in the year before the announcement of the GQ road network upgrades. Regressions include firm fixed effects, *High BGS* × state × year, and *High BGS* × industry × year fixed effects (in column 3). Standard errors, reported in parentheses, are corrected for heteroscedasticity and autocorrelation and clustered at the city level. All variables are defined in Internet Appendix Table 14. * $p < .1$; ** $p < .05$; *** $p < .01$. Data source: CMIE Prowess.

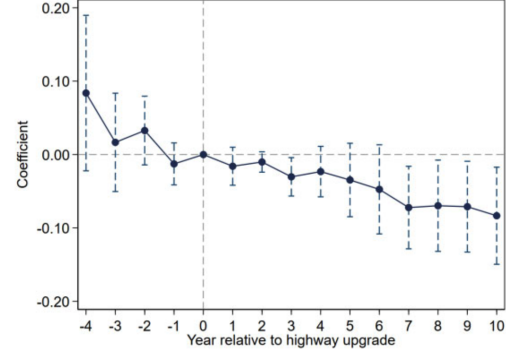


Figure 3
Standalone investment by business group prevalence around GQ upgrades

This figure displays the dynamic coefficients (λ_k) and their corresponding 90% confidence intervals of the differential investment by standalone firms in high- relative to low-business-group share regions around the upgrade of the GQ road network. We estimate a fully dynamic specification that allows us to capture the dynamics of standalone firm investment relative to the year of commencement of the GQ upgrade. Specifically, we estimate the following equation: $Investment_{ijct} = \alpha_i + \sum_{k=-1}^{-4} \mu_k + \sum_{k=1}^{10} \mu_k + \sum_{k=-1}^{-4} \lambda_k \times HighBGS + \sum_{k=1}^{10} \lambda_k \times HighBGS + \omega_{jt} + HighBGS \times \theta_{jt} + \epsilon_{ijct}$. All coefficients are plotted relative to investment at $k=0$, which is normalized to zero. *High BGS* is an indicator variable set to 1 if the share of group assets is in the top quartile of the distribution. The sample is restricted to firms along the GQ network and covers the window of $[-4, +10]$ around the commencement of the upgrade.

5.6 Figure 3: Dynamic standalone investment around GQ upgrades

Read:

- The pre-treatment coefficients are close to zero, which supports the no-differential-pretrends assumption.
- The post-treatment coefficients turn negative and stay negative for many years, with the gap becoming especially visible from about year 2 onward.
- The persistence matters. This is not just a short-run timing shift in investment; it looks like a durable difference in how standalones respond when groups are locally important.

5.7 Table 4: Pre-GQ firm and regional characteristics

Read:

- Pre-GQ standalone firms in high- and low-business-group-share regions look similar on size, age, cash flow, profitability, investment, debt, and inventory days.
- The one notable difference is that average TFP is higher in high-business-group-share regions, if anything biasing against the paper’s finding of lower investment there.
- At the regional level, there is little evidence of meaningful differences in financial development, road quality, or labor-market frictions.
- So the basic observables do not suggest that the baseline result is just a proxy for preexisting regional disadvantage.

5.8 Table 5: Horse race regressions

Read:

Table 4
Pre-GQ firm and regional characteristics by business group share

	A. Standalone firm characteristics		
	High business group share	Low business group share	High – Low (1)–(2)
	(1)	(2)	(3)
Firm size	5.20 (5.05)	5.49 (5.10)	–0.29 (–0.05)
Firm age	19.42 (14.00)	16.87 (14.00)	2.55 0.00
Cash flow	0.06 (0.06)	0.06 (0.06)	0.00 (0.00)
Profitability	0.08 (0.09)	0.11 (0.11)	–0.03 (–0.02)
Investment	0.41 (0.40)	0.47 (0.49)	–0.06 (–0.09)
Debt	0.25 (0.25)	0.28 (0.24)	–0.02 (0.01)
Total factor productivity	3.46 (3.18)	2.73 (2.73)	0.73** 0.45
Days sales of inventory	80.5 (78.8)	86.7 (90.0)	–6.2 –11.2
B. Regional characteristics			
1. Financial development			
Bank branches (per 100,000)	4.95 (4.04)	4.28 (4.06)	0.67 0.26
Fraction of listed firms	0.70 (0.72)	0.73 (0.74)	–0.03 –0.01
2. Physical infrastructure			
Bad Roadways (rating)	6.92 (7.00)	6.77 (7.00)	0.15 0.00
Obstacle to growth, transport (1=yes)	0.26 (0.00)	0.34 (0.00)	–0.08 0.00
3. Labor market conditions			
Labor constraint in contracting (1=yes)	0.07 (0.00)	0.07 (0.00)	0.00 0.00
Obstacle to growth, labor (1=yes)	0.36 (0.00)	0.26 (0.00)	0.09 0.00
Time to fill manager vacancy (weeks)	3.97 (2.00)	3.20 (2.00)	0.77 0.00

This table compares the means (and medians in parentheses) of firm and regional characteristics as a function of the prevalence of business groups in the local area. Panel A presents firm characteristics, while panel B presents regional characteristics. Specifically, panel A displays the means (medians) for standalone firms 1 year before the commencement of GQ upgrades. Column 1 displays mean (and median) for high-business-group share regions while column 2 displays mean (and median) for low-business-group share regions. Column 3 tests the difference in means (medians). *High BGS* is an indicator variable set to 1 if the share of assets of group-affiliated firms from that city is in the top quartile in the year before the GQ road network upgrades. The firm characteristics we focus on are: *Firm size*, *Firm age*, *Cash flow*, *Profitability*, *Investment*, *Debt*, *Total factor productivity*, and *Days sales of inventory*. The regional characteristics we focus on are: *Bank branches*, *Fraction of listed firms*, *Bad roadways*, *Obstacle to growth, transport*, *Labor constraint in contracting*, *Obstacle to growth, labor*, *Time to fill manager vacancy*, and *Average time to fill skilled worker vacancy*. All variables are defined in [Internet Appendix Table 14](#). * $p < .1$; ** $p < .05$; *** $p < .01$. Data sources: CMIE Prowess, Reserve Bank of India, and World Bank Enterprise Survey.

Table 5
Horse Race Regressions

Dependent variable	Investment				
	A. Definition using all firms				
	(1)	(2)	(3)	(4)	(5)
<i>PostGQ</i>	0.039 (0.028)	0.046* (0.027)	0.020 (0.042)	0.002 (0.035)	–0.028 (0.043)
<i>PostGQ</i> × <i>High BGS</i>	–0.038*** (0.012)	–0.084*** (0.029)	–0.106*** (0.029)	–0.105*** (0.041)	–0.170*** (0.058)
<i>PostGQ</i> × <i>High Listed Share (all firms)</i>		–0.216** (0.095)			–0.262*** (0.083)
<i>PostGQ</i> × <i>High Firm Age (all firms)</i>			0.048 (0.054)		0.092 (0.056)
<i>PostGQ</i> × <i>High Firm TFP (all firms)</i>				0.006 (0.025)	0.012 (0.025)
Fixed effects:					
Firm	Yes	Yes	Yes	Yes	Yes
High BGS × industry × year	Yes	Yes	Yes	Yes	Yes
High BGS × state × year	Yes	Yes	Yes	Yes	Yes
Adjusted- R^2	0.70	0.70	0.70	0.66	0.66
Observations	15,842	15,842	15,842	11,520	11,520
Sample: Standalone firms	Yes	Yes	Yes	Yes	Yes
B. Definition using standalone firms					
	(1)	(2)	(3)	(4)	(5)
<i>PostGQ</i>	0.039 (0.028)	0.061** (0.030)	0.005 (0.059)	0.007 (0.036)	0.105* (0.052)
<i>PostGQ</i> × <i>High BGS</i>	–0.038*** (0.012)	–0.099*** (0.032)	–0.095*** (0.029)	–0.110** (0.041)	–0.116** (0.044)
<i>PostGQ</i> × <i>High Listed Share (standalones)</i>		–0.065** (0.025)			–0.009 (0.055)
<i>PostGQ</i> × <i>High Firm Age (standalones)</i>			0.051 (0.055)		0.075 (0.088)
<i>PostGQ</i> × <i>High Firm TFP (standalones)</i>				–0.026 (0.031)	–0.060 (0.036)
Fixed effects:					
Firm	Yes	Yes	Yes	Yes	Yes
High BGS × industry × year	Yes	Yes	Yes	Yes	Yes
High BGS × state × year	Yes	Yes	Yes	Yes	Yes
Adjusted- R^2	0.70	0.70	0.70	0.66	0.66
Observations	15,842	15,842	15,842	11,520	11,520
Sample: Standalone firms	Yes	Yes	Yes	Yes	Yes

This table reports estimates from horse race regressions that relate business group prevalence to standalone firms' investment. We consider the following covariates: *Listed share*, *Firm age*, *Firm size*, and *TFP*. For each covariate, we define an indicator variable, which is set to 1 if the specific characteristic of firms from that city is in the top quartile in the year before the announcement of the GQ road network upgrades. Panel A defines the indicator based on all firms while panel B defines the indicator using only standalone firms. *PostGQ* is an indicator variable taking the value of 1 for all years including and after the GQ upgrade in the city. *High BGS* is an indicator variable set to 1 if the share of assets of group-affiliated firms from that city is in the top quartile in the year before the announcement of the GQ road network upgrades. All regressions include firm fixed effects, *High BGS* × state × year, and *High BGS* × industry × year fixed effects. As TFP is estimated for manufacturing firms, columns 4 and 5 restrict the sample to these industries. Standard errors, reported in parentheses, are corrected for heteroscedasticity and autocorrelation and clustered at the city level. All variables are defined in [Internet Appendix Table 14](#). * $p < .1$; ** $p < .05$; *** $p < .01$. Data source: CMIE Prowess.

- The paper interacts PostGQ with other plausible regional drivers such as high listing share, high firm age, and high TFP.
- Across these specifications, the coefficient on $\text{PostGQ} \times \text{HighBGS}$ remains negative and significant, with magnitudes often even larger than in the baseline.
- This is an important credibility table because it says the business-group-share effect is not being soaked up by other obvious regional characteristics.

5.9 Table 6: Investment by firm type around GQ upgrades

Read:

- For **all firms** together, $\text{PostGQ} \times \text{HighBGS}$ is small and insignificant.
- For **standalone firms**, the interaction is strongly negative: -0.038 .
- For **group-affiliated firms**, the interaction is positive (about 0.009, marginally significant).
- The table suggests a composition effect: aggregate investment is not lower in high-business-group-share regions because group-affiliated firms make up for the weak response of standalones.

Table 6
Investment by firm type around GQ upgrades

Dependent variable	Investment		
	All firms (1)	Standalone (2)	Business Group (3)
Sample			
<i>PostGQ</i>	0.037* (0.021)	0.039 (0.028)	0.024 (0.057)
<i>PostGQ</i> $\times$ <i>High BGS</i>	-0.007 (0.005)	-0.038*** (0.012)	0.009* (0.005)
Fixed effects:			
Firm	Yes	Yes	Yes
High BGS $\times$ industry $\times$ year	Yes	Yes	Yes
High BGS $\times$ state $\times$ year	Yes	Yes	Yes
Adjusted- R^2	0.69	0.70	0.65
Observations	24,319	15,842	8,102

This table reports estimates from regressions relating the prevalence of business groups to investment around GQ road network upgrade for different sample of firms. Column 1 focuses on all firms while column 2 (column 3) focuses on the sample of standalone (group-affiliated) firms. Across all columns, the dependent variable is *Investment*. *PostGQ* is an indicator variable taking the value of 1 for all years including and after the GQ upgrade in the city. *High BGS* is an indicator variable set to 1 if the share of assets of group-affiliated firms from that city is in the top quartile in the year before the announcement of the GQ road network upgrades. All regressions include firm fixed effects, *High BGS* $\times$ state $\times$ year, and *High BGS* $\times$ industry $\times$ year fixed effects. Standard errors, reported in parentheses, are corrected for heteroscedasticity and autocorrelation and clustered at the city level. All variables are defined in [Internet Appendix Table 14](#). * $p < .1$; ** $p < .05$; *** $p < .01$. Data source: CMIE Prowess.

Table 7
Aggregate bank lending around GQ upgrades: District-level evidence

Dependent variable	Log (1+credit)			
	(1)	(2)	(3)	(4)
<i>PostGQ</i>	0.120** (0.047)	0.125*** (0.046)	0.115** (0.052)	0.104** (0.050)
<i>PostGQ</i> $\times$ <i>High BGS</i>			0.032 (0.095)	0.128 (0.114)
Fixed effects:				
District	Yes	Yes	Yes	Yes
Year	Yes	No	Yes	No
State $\times$ year	No	Yes	No	Yes
Adjusted- R^2	0.97	0.98	0.97	0.98
Observations	6,862	6,862	6,862	6,862

This table reports estimates from regressions relating the effect of business group prevalence on overall district-level lending. The dependent variable, *Log (1+credit)*, is defined as the natural logarithm of 1 plus the total credit disbursed, adjusted for inflation, in a district-year. *PostGQ* is an indicator variable taking the value of 1 for all years including and after the GQ upgrade in the city. *High BGS* is an indicator variable set to 1 if a district consists of more than three cities with the share of assets of group-affiliated firms from that city in the top quartile in the year before the GQ road network upgrades. All regressions include district fixed effects. Additionally, specifications in columns 2 and 4 include state $\times$ year fixed effects to control for local macroeconomic confounds, while columns 1 and 3 include year fixed effects to control for time trends. Standard errors, reported in parentheses, are corrected for heteroscedasticity and autocorrelation and clustered at the district level. All variables are defined in [Internet Appendix Table 14](#). * $p < .1$; ** $p < .05$; *** $p < .01$. Data source: Reserve Bank of India.

5.10 Table 7: Aggregate district-level bank lending

Read:

- District credit rises after GQ, with a PostGQ coefficient around 0.12 in logs.
- But the interaction with HighBGS is insignificant.
- So the total amount of bank lending does not differ much across high- and low-business-group-share regions. The mechanism must therefore be about *who gets the credit*, not whether the region gets more credit overall.

5.11 Table 8: Loan-level bank-lending mechanism

Read:

- Banks with greater preexisting exposure to group-affiliated firms reduce their lending to standalone firms after GQ.

- The coefficient on GroupExposure is about -0.185 to -0.191 , statistically significant in both specifications.
- The paper interprets this as roughly an 18.5 to 19.1 percentage point lower loan-growth rate for a one-standard-deviation increase in bank exposure to group firms.
- This is the strongest direct evidence for the financing channel.

Table 8
Loan-level regressions: Bank lending to standalone firms

Dependent variable	Δ Log loan size	
	(1)	(2)
<i>Group exposure</i>	-0.185^{**} (0.071)	-0.191^{**} (0.072)
Firm fixed effects	Yes	Yes
Weighted by firm size	No	Yes
Adjusted- R^2	0.432	0.431
Number of loans	163	163
Sample: Standalone firms	Yes	Yes

The table reports estimates from regressions that examine changes in bank lending for the set of standalone firms borrowing from multiple banks around the upgrade of GQ road network. All loans are time-collapsed into a single pre- and post-period of 5 years around the start of the GQ upgrade. The sample includes standalone firms that borrow from multiple banks. The dependent variable, ΔL_{it} , is the change in the average loan amount to standalone firms 5 years after relative to 5 years before the GQ upgrade. The independent variable, *Group exposure*, is defined for each bank as the total lending to group affiliates before the start of the GQ upgrade, which we standardize to have a mean of zero and a standard deviation of one, for ease of interpretation of the coefficient estimate. The empirical specification includes firm fixed effects that control for the firm-specific credit demand. Standard errors, reported in parentheses, are corrected for heteroscedasticity and autocorrelation and clustered at the bank level. All variables are defined in Internet Appendix Table 14. * $p < .1$; ** $p < .05$; *** $p < .01$. Data source: Ministry of Corporate Affairs.

Table 9
Mechanism: Crowding out demand for standalone firms' output

Dependent variable	Investment	
	Manufacturing	High-exporting
Sample industries	(1)	(2)
<i>PostGQ</i>	-0.008 (0.030)	0.013 (0.044)
<i>PostGQ</i> $\times$ <i>High BGS</i>	-0.081^{***} (0.000)	-0.049^* (0.028)
Fixed effects:		
Firm	Yes	Yes
High BGS $\times$ industry $\times$ year	No	Yes
High BGS $\times$ state $\times$ year	Yes	Yes
Adjusted- R^2	0.66	0.69
Observations	11,521	9,081
Sample: Standalone firms	Yes	Yes

The table reports estimates from regressions that rule out the alternative mechanism whereby group-affiliated firms crowd out demand for standalone firms' output in product markets. Column 1 focuses on the subsample of firms operating in manufacturing industries while column 2 focuses on the subsample of firms operating in "high exporting" industries. We define "high-exporting" industries as those with a ratio of export earnings to sales above the median before the GQ upgrade. *PostGQ* is an indicator variable taking the value of 1 for all years including and after the GQ upgrade in the city. *High BGS* is an indicator variable set to 1 if the share of assets of group-affiliated firms from that city is in the top quartile in the year before the announcement of the GQ road network upgrades. All regressions include firm fixed effects and *High BGS* $\times$ state $\times$ year fixed effects, while column 2 additionally includes *High BGS* $\times$ industry $\times$ year fixed effects. Standard errors, reported in parentheses, are corrected for heteroscedasticity and autocorrelation and clustered at the city level. All variables are defined in Internet Appendix Table 14. * $p < .1$; ** $p < .05$; *** $p < .01$. Data source: CMIE Prowess.

5.12 Table 9: Product-market crowd-out test

Read:

- If business groups crowd out local product demand, the main effect should weaken in sectors that do not rely much on local demand.
- It does *not* weaken. The interaction is -0.081 in manufacturing and -0.049 in high-exporting industries.
- Because these industries sell into broader markets, the table argues against a simple local-demand crowd-out story.

5.13 Table 10: Rival investment opportunities test

Read:

- The paper studies products reserved for smaller firms, where large business-group affiliates are less likely to be allowed to enter.
- Even there, the interaction remains strongly negative: about -0.152 , -0.171 , and -0.236 across the reported specifications.
- That weakens the alternative story that group affiliates simply seize the same high-return projects faster than standalones.

5.14 Internet Appendix highlights

Read:

- **Political connections.** In infrastructure-related industries, results become suggestive when business-group prevalence is redefined using only the 25 largest groups, consistent with political connections mattering in procurement-heavy sectors.
- **Alternative definitions.** The baseline survives continuous business-group-share measures, sales-based concentration measures, and quantile-based alternative cutoffs.
- **Exit and composition.** Dropping likely exits or firms with sharp negative sales growth does not eliminate the result, and differential M&A or entry does not explain the pattern.
- **Staggered-DID robustness.** The appendix also confirms the results with the Borusyak-Jaravel-Spiess imputation estimator.

6 Short summary

- **Conclusion.** Business groups have spillover effects on non-affiliated firms: when local investment opportunities improve, standalones invest less in areas where business groups are more prevalent.
- **Intuition.** The most convincing mechanism is that banks tilt scarce lending capacity toward group-affiliated firms, so group expansion crowds out standalone financing.
- **Empirical lesson.** The paper shows how to study the economywide role of business groups without directly comparing group affiliates and standalones. The key is to compare standalones across regions around a common shock.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- **Main finding.** Standalone firms' investment response to the GQ opportunity shock is significantly lower in high-business-group-share regions.
- **Magnitude.** In the preferred specification, the incremental investment response is lower by about 3.8 percentage points, large relative to the average investment rate of standalones.
- **Why it is credible.** The paper validates GQ as a real opportunity shock, shows no pretrends, compares only eventual GQ locations, and extensively rules out observable firm and regional confounds.
- **Mechanism.** Aggregate bank lending rises similarly across regions, but banks with stronger prior exposure to group affiliates reduce lending to standalones. That is exactly what a credit-reallocation story predicts.
- **What does not seem to drive the result.** Simple local-demand crowd out and rival-opportunity stories are weakened by the manufacturing, exporting, and reserved-product tests.
- **Broader message.** Business groups matter not only because of internal capital markets inside the group, but because they can reshape financing conditions for outsiders.

7.2 Economic intuition

- **Business groups as favored borrowers.** Banks may prefer lending to group affiliates because affiliates enjoy intragroup support, have broader banking relationships, or generate more cross-selling opportunities.
- **Common shock, unequal access.** Once a positive investment opportunity arrives, the issue is not that standalones do not want to invest. It is that they cannot finance investment as easily when competing with business groups.
- **Composition rather than aggregate collapse.** High-business-group-share regions do not necessarily invest less overall. Instead, the composition of investment shifts toward group-affiliated firms and away from standalones.
- **Policy ambiguity.** That shift could be harmful if it reflects distorted credit allocation, but it could be efficient if group affiliates are better uses of capital. The paper documents the spillover, not the welfare sign.

7.3 Takeaway

This is a very clean paper on the external effects of business groups. Its empirical idea is simple and powerful: use a common investment-opportunity shock and compare standalone firms across places with different group prevalence. The result is that business groups do not just affect firms inside the group; they also shape which outside firms get to expand when new opportunities arise, with bank-credit reallocation playing the leading role.